

MMT2:

Da. proof Assume g and f
are continuous functions.

Let $\epsilon > 0$.

We are given g is continuous
at $f(x_0)$, so by the ϵ - δ
definition of continuity,

$$\exists \delta_1 : |y - f(x_0)| < \delta_1 \Rightarrow |g(y) - g(f(x_0))| < \epsilon. \quad y \in (c, d) \quad (A)$$

Similarly for f , it is continuous at x_0 .

$$\exists \delta_2 : |x - x_0| < \delta_2 \Rightarrow |f(x) - f(x_0)| < \delta_1 \quad x \in (a, b) \quad (B)$$

~~so set~~

$\therefore$ If $|x - x_0| < \delta_2$ holds,

we can set $y = f(x)$,

thus $|f(x) - f(x_0)| < \delta_1$

$\Rightarrow |g(f(x)) - g(f(x_0))| < \epsilon$ from (A)

Thus the composition function

$g \circ f : (a, b) \rightarrow \mathbb{R}$ is continuous $\square$

1) b. ~~proof~~ Assume f and g are continuous functions

~~suppose f is continuous at x_0~~
~~and $(f(x_n))_{n \rightarrow \infty} \rightarrow f(x_0)$~~

suppose (x_n) is an arbitrary sequence such that $(x_n) \rightarrow x_0$ ($x_0 \in (a, b)$)
then by definition of continuity,
 $f(x_n) \rightarrow f(x_0)$. ($f(x_0) \in (c, d)$)
(A)

Now, suppose g is continuous at $f(x_0)$. then, ~~by~~ (A) yields.
~~that~~ $g(f(x_n)) \rightarrow g(f(x_0))$

Since $x_n \rightarrow x_0 \Rightarrow g \circ f(x_n) \rightarrow g \circ f(x_0)$
 $g \circ f$ is continuous $\square$

$$2) L(f, P_n) = \sum_{i=0}^{n-1} (r_{i+1} - r_i) \inf_{x \in [r_i, r_{i+1}]} f(x)$$

$$= \sum_{i=0}^{n-1} \frac{1}{n} \cdot \left(\frac{i}{n}\right)^3$$

$$= \frac{1}{n^4} \sum_{i=0}^{n-1} i^3$$

$$= \frac{1}{n^4} \cdot \left(\frac{(n-1)^2 n^2}{4} \right)$$

$$= \frac{n^3 - n^2}{4n^4} = \frac{(n-1)^2}{4n^2}$$

$$= \frac{n-1}{4n^2} = \left(\frac{n-1}{2n} \right)^2$$

$$= \left(\frac{1}{2} - \frac{1}{2n} \right)^2 \xrightarrow{n \rightarrow \infty} \frac{1}{4}$$

$$U(f, P_n) = \sum_{i=0}^n (r_{i+1} - r_i) \sup_{x \in [r_i, r_{i+1}]} f(x)$$

$$= \sum_{i=0}^n \frac{1}{n} \cdot \left(\frac{i+1}{n}\right)^3$$

$$= \frac{1}{n^4} \sum_{i=0}^n i^3$$

$$= \frac{1}{n^4} \cdot \frac{(n+1)^2 n^2}{4}$$

$$= \frac{(n+1)^2}{4n^2} = \left(\frac{1}{2} + \frac{1}{2n} \right)^2 \xrightarrow{n \rightarrow \infty} \frac{1}{4}$$

$$\int_0^1 x^3 dx = \int_0^1 x^3 dx = \frac{1}{4}$$

$$\therefore \int_0^1 x^3 dx = \frac{1}{4}$$

$$3) a. \sum_{n=1}^{\infty} \frac{2}{5n+6}$$

$$= \frac{2}{5+6} + \frac{2}{10+6} + \frac{2}{15+6} \dots$$

$$= \frac{2}{11} + \frac{2}{16} + \frac{2}{21} + \dots$$

Comparing the n^{th} term with the n^{th} term of the harmonic series, ~~b_n~~ $(b_n) = (\frac{1}{n})$

$$\text{then, } \frac{2}{11}(b_n) \leq \left(\frac{2}{5n+6} \right) (\forall n \in \mathbb{Z}^+)$$

Since (b_n) diverges, the series $\sum_{n=1}^{\infty} \frac{2}{5n+6}$ diverges by

the comparison test.

3) b. $\sum_{n=1}^{\infty} \frac{4}{5n^2-4}$

As $n \rightarrow \infty$

$5n^2-4$ is close to $5n^2$, so

$\frac{4}{5n^2-4}$ is close to $\frac{4}{5n^2}$.

Select the series $\sum_{n=1}^{\infty} \frac{4}{n^2}$.

which is convergent, as

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ is the p-series where $p = 2 > 1$.

$$\forall n \in \mathbb{Z}^+, \frac{4}{n^2} \geq \frac{4}{5n^2-4}$$

so by the comparison test,

$\sum_{n=1}^{\infty} \frac{4}{5n^2-4}$ is convergent.

3) a. $\sum_{n=5}^{\infty} \frac{1}{n-4}$

since $n-4 \leq n, \forall n \geq 5$

$$\frac{1}{n-4} \geq \frac{1}{n}, \forall n \geq 5.$$

We know $\sum_{n=5}^{\infty} \frac{1}{n} = \sum_{n=1}^{\infty} \frac{1}{n} - \underbrace{\sum_{n=1}^4 \frac{1}{n}}_{\text{finite}}$
 is the harmonic series, which is divergent, so by the comparison test, $\sum_{n=5}^{\infty} \frac{1}{n-4}$ is divergent.

$$k=1$$

$$3)d. \sum_{n=1}^{\infty} \frac{1}{(n+1)^3}$$

As $n \rightarrow \infty$, $(n+1)^3$ gets close to n^3 , so $\frac{1}{(n+1)^3}$ gets close to $\frac{1}{n^3}$.

We know $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is the p -series

where $p=3>1$, so it converges.

By the comparison test,

$$\frac{1}{n^3} \geq \frac{1}{(n+1)^3} \quad (\forall n \geq 1)$$

so $\sum_{n=1}^{\infty} \frac{1}{(n+1)^3}$ converges.

$$4) a. \sum_{n=1}^{\infty} \frac{1}{n+2}$$

As $n \rightarrow \infty$, $\frac{1}{n+2}$ gets close to $\frac{1}{n}$.

We know $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent as it

is the harmonic series.

Using the limit comparison test, $b_n = \frac{1}{n}$ and $a_n = \frac{1}{n+2}$,

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n+2}}{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{n+2}{n}$$

$$= \lim_{n \rightarrow \infty} 1 + \frac{2}{n}$$

$$= 1$$

Therefore the series is divergent.

$$4)b. \sum_{n=1}^{\infty} \frac{1}{2n^3+9}$$

As $n \rightarrow \infty$ $\frac{1}{2n^3+9}$ gets close to

$$\frac{1}{2n^3}.$$

We know $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is the p -series

where $p=3>1$, so it converges.

Using the limit comparison test, $c_n = \frac{1}{n^3}$, $a_n = \frac{1}{2n^3+9}$.

$$\lim_{n \rightarrow \infty} \frac{a_n}{c_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2n^3+9}}{\frac{1}{n^3}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^3 + \frac{9}{2} - \frac{9}{2}}{2n^3+9}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} - \frac{9}{2(2n^3+9)}$$

$$= \frac{1}{2}$$

So, the series is convergent.

$$4. c. \sum_{n=1}^{\infty} \frac{1}{3n^2+4n-2}$$

As $n \rightarrow \infty$, $3n^2+4n-2$ gets closer to $3n^2$, so

$\frac{1}{3n^2+4n-2}$ gets closer to $\frac{1}{3n^2}$.

We know $\sum_{n=1}^{\infty} \frac{1}{n^2}$ ~~con~~ is the p-series where $p=2 > 1$, so it converges.

Using the limit comparison test, $(a_n) = \left(\frac{1}{3n^2+4n-2}\right)$ and

$$(c_n) = \left(\frac{1}{n^2}\right).$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{c_n} = \lim_{n \rightarrow \infty} \frac{1}{3n^2+4n-2} \cdot n^2$$

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{a_n}{c_n} &= \lim_{n \rightarrow \infty} \frac{\frac{1}{3n^2+4n-2}}{\frac{1}{n^2}} \\
 &= \lim_{n \rightarrow \infty} \frac{n^2}{3n^2+4n-2} \times \frac{\frac{1}{n^2}}{\frac{1}{n^2}} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{3 + \frac{4}{n} - \frac{2}{n^2}} \\
 &= \frac{1}{3}
 \end{aligned}$$

Therefore the ~~series~~ given series is convergent.

4)d. $\sum_{n=1}^{\infty} \frac{1}{2^n}$ for $2 > 2$.

$$\frac{1}{2^n} = (2)^{-n} = \left(\frac{1}{2}\right)^n$$

Since $2 > 2$, $\frac{1}{2} < \frac{1}{2} < 1$.

~~Let $(a_n) = \left(\frac{1}{2^n}\right)$, $(c_n) = \left(\frac{1}{2^n}\right)$~~

We know $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ is a convergent

geometric series as $r = \frac{1}{2} < 1$.

Using the limit comparison test.

$(a_n) = \left(\frac{1}{2^n}\right)$ and $(c_n) = \left(\frac{1}{2^n}\right)$

~~Let~~

$$\lim_{n \rightarrow \infty} \frac{a_n}{c_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2^n}}{\frac{1}{2^n}}$$

$$= \lim_{n \rightarrow \infty} \frac{2^n}{2^n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{2}{2}\right)^n$$

Since $2 < 2$, $\frac{2}{2} < 1$.

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{2}{2}\right)^n = \frac{1}{1 - \frac{2}{2}}$$

$$= \frac{2}{2-2}$$

So, the given series converge.

5) a. $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

$$u_n = \frac{n^2}{2^n}, \quad u_{n+1} = \frac{(n+1)^2}{2^{n+1}} \quad \forall n \geq 1$$

Therefore,

$$\frac{u_{n+1}}{u_n} = \frac{\frac{(n+1)^2}{2^{n+1}}}{\frac{n^2}{2^n}} = \frac{2^n (n+1)^2}{n^2 (2^{n+1})}$$

$$= \frac{n^2 + 2n + 1}{2n^2}$$

$$= \frac{1}{2} + \frac{1}{n} + \frac{1}{2n^2} \rightarrow \frac{1}{2} < 1$$

So, by the D'Alembert ratio test, the given series is absolutely convergent.

$$5) b. \sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$$

$$u_n = \frac{(n!)^2}{(2n)!}$$

$$u_{n+1} = \frac{((n+1)!)^2}{(2(n+1))!} \quad \text{Ans 1}$$

$$\frac{u_{n+1}}{u_n} = \frac{\frac{((n+1)!)^2}{(2n+2)!}}{\frac{(n!)^2}{(2n)!}} = \frac{(n+1)^2 \cancel{(n!)^2} (2n)!}{(2n+2)! \cancel{(n!)^2}}$$

$$= \frac{(n+1)^2 (2n)!}{(2n+2)!} = \frac{(n+1)^2 (2n)!}{(2n+2)(2n+1)(2n)!} \cdot \frac{\frac{1}{n^2}}{\frac{1}{n^2}}$$

$$= \frac{(1 + \frac{1}{n})^2}{(2 + \frac{2}{n})(2 + \frac{1}{n})}$$

$$\rightarrow \frac{1}{2 \cdot 2} = \frac{1}{4} < 1$$

Therefore by the D'Alembert's limit Ratio test, the given series converge.

$$5) c. \sum_{n=1}^{\infty} \frac{3^n}{n^3 2^n}$$

$$a_n = \frac{3^n}{n^3 2^n} \quad a_{n+1} = \frac{3^{n+1}}{(n+1)^3 2^{n+1}} \quad \forall n \geq 1$$

$$\frac{a_{n+1}}{a_n} = \frac{\frac{3^{n+1}}{(n+1)^3 2^{n+1}}}{\frac{3^n}{n^3 2^n}}$$

$$= \frac{3^{n+1} n^3 2^n}{3^n (n+1)^3 2^{n+1}}$$

$$= \frac{3 n^3}{2 (n+1)^3} \cdot \frac{\frac{1}{n^3}}{\frac{1}{n^3}}$$

$$= \frac{3}{2 \left(1 + \frac{1}{n}\right)^3} \rightarrow \frac{3}{2} > 1$$

So, the given series diverges by the D'Alembert ratio test

$$5) d. \sum_{n=1}^{\infty} \frac{n!}{n^n}$$

$$a_n = \frac{n!}{n^n} \quad a_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}} \quad \forall n \geq 1$$

$$\frac{a_{n+1}}{a_n} = \frac{\frac{(n+1)!}{(n+1)^{n+1}}}{\frac{n!}{n^n}}$$

$$= \frac{(n+1)! \cdot n^n}{n! \cdot (n+1)^{n+1}}$$

$$= \frac{\cancel{(n+1)} \cdot \cancel{(n!)} \cdot n^n}{n! \cdot (n+1) \cdot (n+1)^n}$$

$$= \frac{n^n}{(n+1)^n}$$

$$= \left(\frac{n}{n+1} \right)^n \cdot \frac{\left(\frac{1}{n} \right)^n}{\left(\frac{1}{n} \right)^n}$$

$$= \left(\frac{1}{1 + \frac{1}{n}} \right)^n$$

$$= \frac{1}{\left(1 + \frac{1}{n} \right)^n} \rightarrow \frac{1}{e} < 1$$

using $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n \rightarrow e$

Therefore, the given series converges by the D'Alembert ratio test.

$$6) \sum_{n=1}^{\infty} \frac{1}{e^n}$$

Let $f(x) = \frac{1}{e^x}$ which is continuous, decreasing and positive for $\forall x \in [1, \infty)$.

Thus, considering the improper integral, we have,

$$\begin{aligned} \int_1^{\infty} \frac{1}{e^x} dx &= \lim_{b \rightarrow \infty} \int_1^b \frac{1}{e^x} dx \\ &= \lim_{b \rightarrow \infty} \left[-\frac{1}{e^x} \right]_1^b \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{e^b} + \frac{1}{e} \right) \\ &= \frac{1}{e} \end{aligned}$$

$$\begin{aligned} \int_1^{\infty} \frac{dx}{e^x} &= \lim_{b \rightarrow \infty} \int_1^b \frac{dx}{e^x} \\ &= \lim_{b \rightarrow \infty} \left[-\frac{1}{e^x} \right]_1^b \\ &= \lim_{b \rightarrow \infty} \left(\frac{1}{e} - \frac{1}{e^b} \right) \\ &= \frac{1}{e} \end{aligned}$$

~~Thus~~ $\therefore$ by the integral test, the given series converges.

$$1) \sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}, \alpha > 1$$

$$\text{let } a_n = \frac{1}{n^{\alpha}} \quad a_{n+1} = \frac{1}{(n+1)^{\alpha}} \quad \forall n \geq 1 \quad \forall \alpha > 1$$

$$\frac{a_{n+1}}{a_n} = \frac{\frac{1}{(n+1)^{\alpha}}}{\frac{1}{n^{\alpha}}}$$

$$= \frac{n^{\alpha}}{(n+1)^{\alpha}}$$

$$= \left(\frac{n}{n+1} \right)^{\alpha} = \left(\frac{\frac{1}{n}}{\frac{1}{n} + \frac{1}{n^2}} \right)^{\alpha}$$

$$= \left(\frac{1}{1 + \frac{1}{n}} \right)^{\alpha} \rightarrow 1$$

$$1) \sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}, \alpha > 1$$

let $f: \mathbb{R} \rightarrow \mathbb{R}^+$ be defined as $f(x) = \frac{1}{x^{\alpha}}$ for $\forall \alpha > 1, x \in [1, \infty)$ which is continuous, decreasing and positive.

Thus,

$$\int_1^{\infty} \frac{dx}{x^{\alpha}} = \lim_{b \rightarrow \infty} \int_1^b \frac{dx}{x^{\alpha}}$$

$$= \lim_{b \rightarrow \infty} \left[\frac{1}{x^{\alpha-1}(1-\alpha)} \right]_1^b$$

$$= \lim_{b \rightarrow \infty} \frac{1}{b^{(\alpha-1)}(1-\alpha)} - \frac{1}{1-\alpha}$$

$$= -\frac{1}{1-\alpha} = \frac{1}{\alpha-1}$$

$\therefore$ by the integral test, the series converges.

$$8) a. \sum_{n=1}^{\infty} \frac{1}{3n+2} = \frac{1}{5} + \frac{1}{8} + \dots$$

As $n \rightarrow \infty$, $\frac{1}{3n+2}$ gets close to $\frac{1}{3n}$.

We know the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, so $\sum_{n=1}^{\infty} \frac{1}{3n}$ ~~diverges~~

also diverges.

$$\forall n \geq 1, \frac{1}{3n} \leq \frac{1}{3n+2}$$

by the comparison test, the given series diverges.

$$b. \sum_{n=1}^{\infty} \frac{1}{1+n^2}$$

As $n \rightarrow \infty$, $\frac{1}{1+n^2}$ gets close to $\frac{1}{n^2}$.

We know $\sum \frac{1}{n^2}$ ~~converges~~ is a p series where $p=2 > 1$, so the series converges.

$$\forall n \geq 1, \frac{1}{n^2} \geq \frac{1}{1+n^2}$$

so by the comparison test, the given series converges.

$$c. \sum_{n=1}^{\infty} n! x^n, \quad x \in \mathbb{R}$$

$$\text{let } a_n = n! x^n, \quad a_{n+1} = (n+1)! x^{n+1} \quad (n \geq 1)$$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)! x^{n+1}}{n! x^n}$$

$$= \frac{(n+1)(\cancel{n!}) x^n \cdot x}{(\cancel{n!}) x^n}$$

$$= x(n+1)$$

for $x > 0$,

$$x(n+1) \rightarrow +\infty$$

for $x < 0$,

$$x(n+1) \rightarrow -\infty$$

So, by the D'Alembert ratio test,
the series diverges $\forall x \in \mathbb{R}$.

$$d. \sum_{n=1}^{\infty} \left(\frac{x}{n}\right)^n, \quad x \in \mathbb{R}$$

$$\text{let } a_n = \left(\frac{x}{n}\right)^n, \quad a_{n+1} = \left(\frac{x}{n+1}\right)^{n+1} \quad \forall x \geq 1$$

$$\frac{a_{n+1}}{a_n} = \frac{\left(\frac{x}{n+1}\right)^{n+1}}{\left(\frac{x}{n}\right)^n} = \frac{x}{n+1} \cdot \left(\frac{x}{n+1}\right)^n \cdot \left(\frac{n}{x}\right)^n$$

$$= \frac{x}{n+1} \left(\frac{n}{n+1}\right)^n \cdot \left(\frac{1}{1/n}\right)^n$$

$$= \frac{x}{n+1} \left(\frac{1}{1+\frac{1}{n}}\right)^n$$

$$\forall x \in \mathbb{R}, \frac{x}{n+1} \xrightarrow{n \rightarrow \infty} 0$$

$$\left(\frac{1}{1+\frac{1}{n}} \right)^n \xrightarrow{n \rightarrow \infty} \frac{1}{e}$$

$$\Rightarrow \frac{x}{n+1} \left(\frac{1}{1+\frac{1}{n}} \right)^n \rightarrow 0 < 1$$

So, by D'Alembert's ratio test, the series converges. $\forall x \in \mathbb{R}$.

$$e. \sum_{n=1}^{\infty} \sin \frac{\pi}{n}, \quad 2x < \sin x < x \text{ for } 0 < x < \frac{\pi}{2}$$

$$\Rightarrow \frac{2}{n} < \sin \frac{\pi}{n} < \frac{\pi}{n}$$

$$= \sum_{n=3}^{\infty} \sin \frac{\pi}{n} + \underbrace{\sum_{n=1}^2 \sin \frac{\pi}{n}}_{\text{finite no. of terms, so converges.}}$$

$$\text{We know } \forall n \geq 3, \quad 0 < \frac{\pi}{n} < \frac{\pi}{2}$$

$$\Rightarrow \frac{2}{n} < \sin \frac{\pi}{n} < \frac{\pi}{n}.$$

so ~~since~~ since $\sum_{n=1}^{\infty} \frac{2}{n}$ diverges by the

harmonic series, the given series diverges by the comparison test.

$$8) f. \sum_{n=1}^{\infty} \frac{\sin nx}{n^2}, \quad x \in \mathbb{R}.$$

since $|\sin nx| \leq 1; (\forall x \in \mathbb{R}, n \geq 1)$

$$\left| \frac{\sin nx}{n^2} \right| \leq \frac{1}{n^2}.$$

We know $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a p -series where $p=2 > 1$, so the series converges.

$\therefore$ by the absolute comparison test, the series converges $\forall x \in \mathbb{R}$.

$$9) \sum_{n=1}^{\infty} \frac{(2n)!}{2^{2n}(n!)^2} = \frac{2!}{2^2(1!)^2} + \frac{4!}{2^4(2!)^2} + \frac{6!}{2^6(3!)^2} + \dots$$

$$= \frac{1}{2} + \dots + \frac{3}{8} + \frac{5}{16} + \dots$$

claim $\forall n \geq 1, \frac{1}{2n} \leq \frac{(2n)!}{2^{2n}(n!)^2}$

let $n=1$,

$$\frac{1}{2n} = \frac{1}{2(1)} = \frac{1}{2}$$

$$\frac{(2n)!}{2^{2n}(n!)^2} = \frac{2!}{2^{2(1)}(1!)^2} = \frac{2!}{4} = \frac{1}{2}$$

$\frac{1}{2} \leq \frac{1}{2}$ holds, so true for $n=1$.

Assume true for $n=k$.

i.e. $\frac{1}{2k} \leq \frac{(2k)!}{2^{2k}(k!)^2}$

We need to show that

$$\frac{1}{2(k+1)} \leq \frac{(2(k+1))!}{2^{2(k+1)}((k+1)!)^2}$$

let $n=k+1$,

$$\frac{(2(k+1))!}{2^{2(k+1)}((k+1)!)^2} = \frac{(2k+2)(2k+1)(2k)!}{2^2 \cdot 2^{2k} \cdot (k+1)^2 (k!)^2}$$

$$= \frac{(2k+2)(2k+1)}{4(k+1)^2} \cdot \frac{(2k)!}{2^{2k}(k!)^2} \geq \frac{(2k+2)(2k+1)}{4(k+1)^2} \cdot \frac{1}{2k}$$

by assumption.

$$= \frac{2(k+1)(2k+1)}{4(k+1)^2} \cdot \frac{1}{2k}$$

$$= \frac{2k+1}{4k(k+1)}$$

we need to show

$$\frac{2k+1}{4k(k+1)} \geq \frac{1}{2(k+1)} \quad \forall k \geq 1.$$

$$\Leftrightarrow \frac{2k+1}{2k} \geq 1$$

$$\Leftrightarrow 1 + \frac{1}{2k} \geq 1 \quad \text{which holds } \forall k \geq 1.$$

~~so, if true for $n=k$, it is also true for~~

so, if true for $n=k$, $\frac{1}{2^n} \leq \frac{(2n)!}{2^{2n}(n!)^2}$ also holds for $n=k+1$.

Since true for $n=1$, it is $\therefore$ true for $n \geq 1$.

We know that $\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n}$

which diverges (harmonic series).

Using the comparison test,

$$\text{since } \forall n \geq 1, \frac{1}{2^n} \leq \frac{(2n)!}{2^{2n}(n!)^2}$$

the given series diverges.